

JEE MAIN REVISION MASTER: MATHEMATICAL REASONING OUT OF SYLLABUS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 121 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Mathematical Reasoning - Core Application	121 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Mathematical Reasoning - Core Application (Very High Yield — 121 PYQs)

Core Principle: Key problem pattern in Mathematical Reasoning.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2023 Morning Shift] Q#1 (MCQ): "Negation of $p \wedge (q \wedge \sim(p \wedge q))$ is :..." [Ans: N/A]
- [2023 Evening Shift] Q#1 (MCQ): "The converse of $((\sim p) \wedge q) \Rightarrow r$ is..." [Ans: N/A]
- [2023 Morning Shift] Q#1 (MCQ): "Negation of $(p \Rightarrow q) \Rightarrow (q \Rightarrow p)$ is :..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2023 (Online) 15th April Morning Shift (Q#1)

Negation of $p \wedge (q \wedge \sim(p \wedge q))$ is :

- (A) $(\sim(p \wedge q)) \wedge q$
- (B) $(\sim(p \wedge q)) \vee p$
- (C) $p \vee q$
- (D) $\sim(p \vee q)$

Correct Answer: None

Solution Analysis: The given statement is: $p \wedge (q \wedge \sim(p \wedge q))$ We want to find the negation of this statement. To do that, we negate the whole expression: $\sim(p \wedge (q \wedge \sim(p \wedge q)))$ Now, we will use De Morgan's laws to simplify the expression. De Morgan's laws state: $\sim(A \wedge B) \equiv \sim A \vee \sim B$ $\sim(A \vee B) \equiv \sim A \wedge \sim B$ Applying the first De Morgan's law to the outer conjunction of the negated expression: $\sim p \vee \sim(q \wedge \sim(p \wedge q))$ Now we need to address the inner conjunction $(q \wedge \sim(p \wedge q))$. To simplify this part of the expression, we again apply the first De Morg